

# State Diagrams

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## State Diagram Software

We will be generating state diagrams all semester, which can be done with a variety of different software packages. You can use whatever software package you prefer for this class this semester, though Mermaid will be the only software supported by Dr. Palmeri and the teaching team.

- [Mermaid Markdown](#)
  - Markdown-based diagramming tool that can be integrated into VS Code IDE with extension.
  - Native Quarto Markdown (what is used for class content) rendering without needing to export to a separate image file.
  - Ammenable to version control systems (VCS) since it is text-based.
  - More modern and easier to use than PlantUML.
  - State digram-specific syntax: <https://mermaid.ai/open-source/syntax/stateDiagram.html>
- [PlantUML](#)
  - Text-based diagramming tool with very long history.
  - Can be integrated into VS Code with an extension.
  - A bit outdated, and a tad clunky in default layouts without some “nudging”

- Ammenable to version control systems (VCS) since it is text-based.

**i** Note

Rendering engine requires Java Runtime Engine (JRE), which may have to be installed on your laptop if you do not already have it. If you would like to avoid installing that, you can use the [PlantUML Server](#) option in the PlantUML VS Code extension settings.

- [draw.io](#)
  - Web-based diagramming tool (GUI) that can be integrated into VS Code (Draw.io Integration)
  - Very feature-rich and easy to use.
  - Not text-based, so less VCS friendly.
  - Cannot directly render in Quarto Markdown, so you will need to export your diagrams to an image file (e.g., PNG, SVG).
- [Lucidchart](#)
  - Web-based diagramming tool (GUI)
  - Very feature-rich and easy to use
  - Not text-based, so less VCS friendly.